

Basic Concepts of Measurement Methods

1.1 INTRODUCTION

We make measurements every day. Consider the common measurements illustrated in Figure 1.1. We routinely read the temperature of an outdoor thermometer to choose appropriate clothing for the day. We expect to have exactly 10 gallons or liters of fuel added to our tank when that volume is indicated on a fuel pump. And we expect measuring cups to yield correct quantities of ingredients in cooking. We put little thought into the selection of instruments for these routine measurements. After all, the direct use of the data is clear to us, the type of instruments and techniques are familiar to us, and the outcome of these measurements is not important enough to merit much attention to features like improved accuracy or alternative methods. But when the stakes become greater, the selection of measurement equipment and techniques and the interpretation of the measured data can demand considerable attention. Just contemplate how you might verify that a new engine is built as designed and meets the power and emissions performance specifications required.

But first things first. The objective in any measurement is to answer a question. So we take measurements to establish the value or the tendency of some variable, the results of which are specifically targeted to answer our question. The information acquired is based on the output of the measurement device or system. There are important issues to be addressed to ensure that the output of the measurement device is a reliable indication of the true value of the measured variable. In addition, we must address the following important questions:

1. How can a measurement or test plan be devised so that the measurement provides the unambiguous information we seek?
2. How can a measurement system be used so that the engineer can easily interpret the measured data and be confident in their meaning?

There are procedures that address these measurement questions.

At the onset, we want to stress that the subject of this text is real-life oriented. Specifying a measurement system and measurement procedures represents an open-ended design problem whose outcome will not have one particular solution. That means there may be several approaches to solving a measurement problem, and some will be better than others. This text emphasizes accepted procedures for analyzing a measurement problem to assist in the selection of equipment,

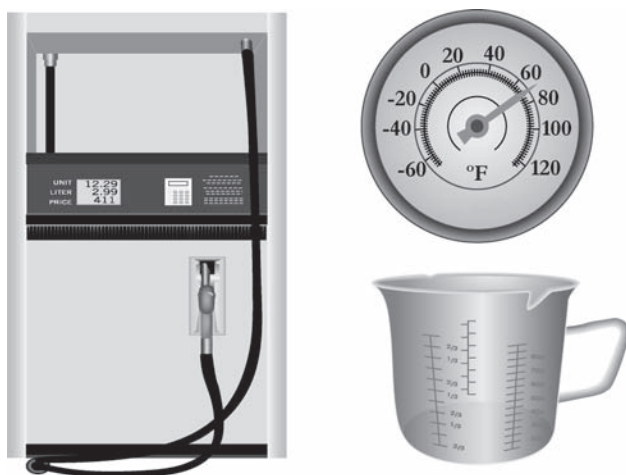


Figure 1.1 Common devices that involve measurements.

methodology, and data analysis to meet the design objectives. Perhaps more than in any other technical field, the approach taken in measurement design and the outcome achieved will often depend on the attention and experience of the designer.

Upon completion of this chapter, the reader will be able to

- identify the major components of a general measurement system, and state the function of each,
- develop an experimental test plan,
- distinguish between random and systematic errors,
- describe and define the various error types,
- define a standard and distinguish among primary, secondary, and transfer standards, and
- clearly delineate defined and derived dimensions in various unit systems.

1.2 GENERAL MEASUREMENT SYSTEM

A *measurement*¹ is an act of assigning a specific value to a physical variable. That physical variable is the *measured variable*. A measurement system is a tool used for quantifying the measured variable. As such, a measurement system is used to extend the abilities of the human senses that, while they can detect and recognize different degrees of roughness, length, sound, color, and smell, are limited and relative; they are not very adept at assigning specific values to sensed variables.

A system is composed of components that work together to accomplish a specific objective. We begin by describing the components that make up a measurement system, using specific examples. Then we will generalize to a model of the generic measurement system.

¹ There are many new engineering measurement terms introduced. A glossary of the italicized terms is located in the back of the text for your reference.

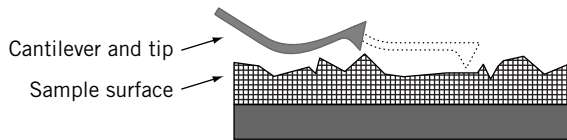


Figure 1.2 Sensor stage of an atomic-force microscope.

Sensor and Transducer

An increasingly important area of scientific inquiry is the characteristics of matter at the nanoscale. Suppose we want to measure the profile of a surface at a nanometer scale. We discover that a small (very small) cantilever beam placed near the surface is deflected by atomic forces. Let's assume for now that they are repulsive forces. If this cantilever is translated over the surface, the cantilever will deflect, indicating the height of the surface. This concept is illustrated in Figure 1.2; the device is called an atomic force microscope. The cantilever beam is a *sensor*, a physical element that employs some natural phenomenon, in this case deflection under the action of a force, to sense the variable being measured, in this case the height of the surface.

So, we have a sensor to measure at the nanometer scale. But we have no means of getting an output from the sensor that we can record. Suppose that the upper surface of the cantilever is reflective, and we shine a laser onto the upper surface, as shown in Figure 1.3. The movement of the cantilever will deflect the laser. Employing a number of light sensors, also shown in Figure 1.3, the deflection of the laser can be sensed and that deflection corresponds to the height of the surface. Together the laser and the light sensors (photodiodes) form the transducer component of the measurement system. A transducer converts the sensed information into a detectable signal. The signal might be mechanical, electrical, optical, or may take any other form that can be meaningfully recorded.

We should note that sensor selection, placement, and installation are particularly important to ensure that the sensor output accurately reflects the measurement objective. The familiar phrase

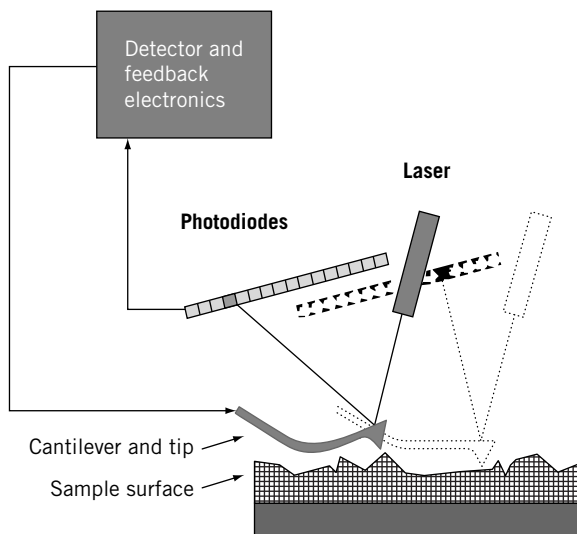


Figure 1.3 Atomic-force microscope with sensor and transducer stages.

associated with hot days, “100°F in the shade” implies a specific sensor placement. Accordingly, the interpretation of all information passed through and indicated by the system depends on what is actually sensed by the sensor. For example, the interpretation of the output of a medical thermometer depends on where its sensor is placed.

Output Stage

The goal of a measurement system is to convert the sensed information into a form that can be easily quantified. Consider a familiar example, the liquid-in-glass bulb thermometer. The liquid contained within the bulb on the common bulb thermometer of Figure 1.4 exchanges energy with its surroundings until the two are in thermal equilibrium. At that point they are at the same temperature. This energy exchange is the input signal to this measurement system. The phenomenon of thermal expansion of the liquid results in its movement up and down the stem, forming an output signal from which we determine temperature. The liquid in the bulb acts as the sensor. By forcing the expanding liquid into a narrow capillary, this measurement system transforms thermal information into a mechanical displacement. Hence, the bulb’s internal capillary design acts as a transducer.

The *output stage* indicates or records the value measured. This might be a simple readout display, a marked scale, or even a recording device such as a computer disk drive. The readout scale of the bulb thermometer in Figure 1.4 serves as the output stage of that measurement system.

It is worth noting that the term “transducer” is also often used in reference to a packaged device, which may contain a sensor, transducer, and even some signal conditioning elements. While such terminology is not true to our presentation, the context in which the term is used prevents ambiguity.

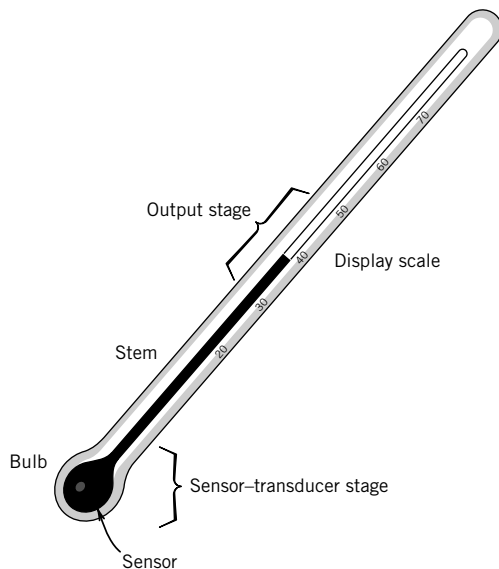


Figure 1.4 Components of bulb thermometer equivalent to sensor, transducer, and output stages.

General Template for a Measurement System

A general template for a measurement system is illustrated in Figure 1.5. Basically such a system consists of part or all of four general stages: (1) sensor–transducer stage, (2) signal-conditioning stage, (3) output stage, and (4) feedback-control stage. These stages form the bridge between the input to the measurement system and the system output, a quantity that is used to infer the value of the physical variable measured. We discuss later how the relationship between the input information, as acquired by the sensor, and the system output is established by a calibration. We have already discussed the sensor–transducer stage, so let’s move on to the signal-conditioning stage.

The *signal-conditioning stage* takes the transducer signal and modifies it to a desired magnitude. This optional intermediate stage might be used to perform tasks such as increasing the magnitude of the signal by amplification, removing portions of the signal through some filtering technique, or providing mechanical or optical linkage between the transducer and the output stage. For example, the translational displacement of a mechanic’s caliper (sensor) is often converted into a rotational displacement of a pointer. This stage can consist of one or more devices, which are often connected in series. For example, the diameter of the thermometer capillary relative to the bulb volume (see Fig. 1.4) determines how far up the stem the liquid moves with increasing temperature. It “conditions” the signal by amplifying the liquid displacement.

In those measurement systems involved in process control, a fourth stage, the *feedback-control stage*, contains a controller that interprets the measured signal and makes a decision regarding the control of the process. This decision results in a signal that changes the process parameter that affects the magnitude of the sensed variable. In simple controllers, this decision is based on the magnitude of the signal of the sensed variable, usually whether it exceeds some high or low set point, a value set by the system operator. For example, a simple measurement system with control stage is a household furnace thermostat. The operator fixes the set point for temperature on the thermostat display, and the furnace is activated as the local temperature at the thermostat, as determined by the

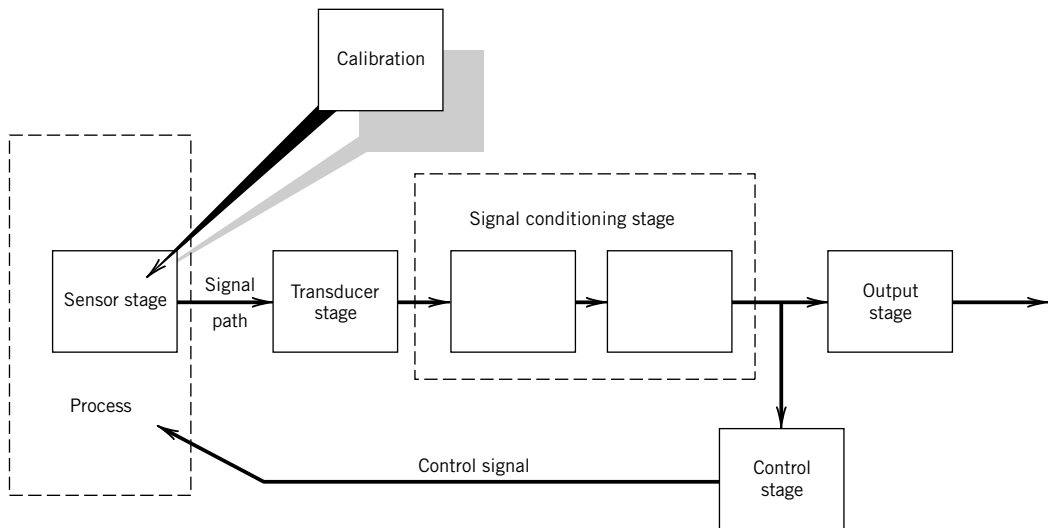


Figure 1.5 Components of a general measurement system.

sensor within the device, rises or falls above or below the set point. In a more sophisticated controller, a signal from a measurement system can be used as an input to an “expert system” controller that, through an artificial intelligence algorithm, determines the optimum set conditions for the process. *Mechatronics* deals with the interfacing of mechanical and electrical components with microprocessors, controllers, and measurements. We will discuss some features of mechatronic systems in detail in Chapter 12.

1.3 EXPERIMENTAL TEST PLAN

An experimental test serves to answer a question, so the test should be designed and executed to answer that question and that question alone. This is not so easy to do. Let’s consider an example.

Suppose you want to design a test to answer the question, “What is the fuel use of my new car?” What might be your test plan? In a test plan, you identify the variables that you will measure, but you also need to look closely at other variables that will influence the result. Two important variables to measure would be distance and fuel volume consumption. Obviously, the accuracy of the odometer will affect the distance measurement, and the way you fill your tank will affect your estimate of the fuel volume. But what other variables might influence your results? If your intended question is to estimate the average fuel usage to expect over the course of ownership, then the driving route you choose would play a big role in the results and is a variable. Only highway driving will impose a different trend on the results than only city driving, so if you do both you might want to randomize your route by using various types of driving conditions. If more than one driver uses the car, then the driver becomes a variable because each individual drives somewhat differently. Certainly weather and road conditions influence the results, and you might want to consider this in your plan. So we see that the utility of the measured data is very much impacted by variables beyond the primary ones measured. In developing your test, the question you propose to answer will be a factor in developing your test plan, and you should be careful in defining that question so as to meet your objective.

Imagine how your test conduct would need to be different if you were interested instead in providing values used to advertise the expected average fuel use of a model of car. Also, you need to consider just how good an answer you need. Is 2 liters per 100 kilometers or 1 mile per gallon close enough? If not, then the test might require much tighter controls. Lastly, as a concomitant check, you might compare your answer with information provided by the manufacturer or independent agency to make sure your answer seems reasonable. Interestingly, this one example contains all the same elements of any sophisticated test. If you can conceptualize the factors influencing this test and how you will plan around them, then you are on track to handle almost any test. Before we move into the details of measurements, we focus here on some important concepts germane to all measurements and tests.

Experimental design involves itself with developing a measurement test plan. A test plan draws from the following three steps:²

1. **Parameter design plan.** Determine the test objective and identify the process variables and parameters and a means for their control. Ask: “What question am I trying to answer? What needs to be measured?” “What variables and parameters will affect my results?”

² These three strategies are similar to the bases for certain design methods used in engineering system design (1).

2. **System and tolerance design plan.** Select a measurement technique, equipment, and test procedure based on some preconceived tolerance limits for error.³ Ask: “In what ways can I do the measurement and how good do the results need to be to answer my question?”
3. **Data reduction design plan.** Plan how to analyze, present, and use the anticipated data. Ask: “How will I interpret the resulting data? How will I use the data to answer my question? How good is my answer? Does my answer make sense?”

Going through all three steps in the test plan before any measurements are taken is a useful habit for a successful engineer. Often, step 3 will force you to reconsider steps 1 and 2! In this section, we focus on the concepts related to step 1 but will discuss and stress all three throughout the text.

Variables

Once we define the question that we want the test to answer, the next step is to identify the relevant process parameters and variables. Variables are entities that influence the test. In addition to the targeted measured variable, there may be other variables pertinent to the measured process that will affect the outcome. All known process variables should be evaluated for any possible cause-and-effect relationships. If a change in one variable will not affect the value of some other variable, the two are considered independent of each other. A variable that can be changed independently of other variables is known as an *independent variable*. A variable that is affected by changes in one or more other variables is known as a *dependent variable*. Normally, the variable that we measure depends on the value of the variables that control the process. A variable may be continuous, in that its value is able to change in a continuous manner, such as stress under a changing load or temperature in a room, or it may be discrete in that it takes on discrete values or can be quantified in a discrete way, such as the value of the role of dice or a test run by a single operator.

The *control* of variables is important. A variable is controlled if it can be held at a constant value or at some prescribed condition during a measurement. Complete control of a variable would imply that it can be held to an exact prescribed value. Such complete control of a variable is not usually possible. We use the adjective “controlled” to refer to a variable that can be held as prescribed, at least in a nominal sense. The cause-and-effect relationship between the independent variables and the dependent variable is found by controlling the values of the independent variables while measuring the dependent variable.

Variables that are not or cannot be controlled during measurement but that affect the value of the variable measured are called *extraneous variables*. Their influence can confuse the clear relation between cause and effect in a measurement. Would not the driving style affect the fuel consumption of a car? Then unless controlled, this influence will affect the result. Extraneous variables can introduce differences in repeated measurements of the same measured variable taken under seemingly identical operating conditions. They can also impose a false trend onto the behavior of that variable. The effects due to extraneous variables can take the form of signals superimposed onto the measured signal with such forms as noise and drift.

³ The tolerance design plan strategy used in this text draws on uncertainty analysis, a form of sensitivity analysis. Sensitivity methods are common in design optimization.

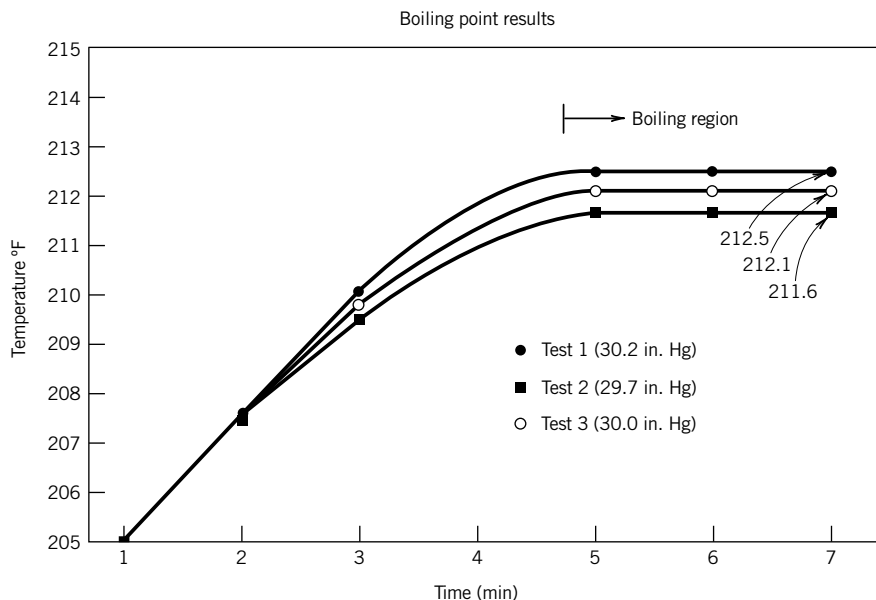


Figure 1.6 Results of a boiling point test for water.

Consider a thermodynamics experiment to establish the boiling point of water. The apparatus for measuring the boiling point might yield the results shown in Figure 1.6 for three test runs conducted on separate days. Notice the different outcome for each test.

Why should the data from three seemingly identical tests show such different results?

Suppose we determine that the measurement system accuracy accounts for only 0.1°F of the test data scatter. So another plausible contributing factor is the effect of an extraneous variable. Indeed, a close examination of the test data shows a measured variation in the barometric pressure, which would affect the boiling temperature. The pressure variation is consistent with the trend seen in the boiling point data. Because the local barometric pressure was not controlled (i.e., it was not held fixed between the tests), the pressure acted as an extraneous variable adding to the differences in outcomes between the test runs. *Control important variables or be prepared to solve a puzzle!*

Parameters

In this text, we define a *parameter* as a functional grouping of variables. For example, a moment of inertia or a Reynolds number has its value determined from the values of a grouping of variables. A parameter that has an effect on the behavior of the measured variable is called a *control parameter*. Available methods for establishing control parameters based on known process variables include similarity and dimensional analysis techniques and physical laws (2–4). A parameter is controlled if its value can be maintained during a set of measurements.

As an example, the flow rate, Q , developed by a fan depends on rotational speed, n , and the diameter, d , of the fan. A control parameter for this group of three variables, found by similarity methods, is the fan flow coefficient, $C_1 = Q/nd^3$. For a given fan, d is fixed (and therefore controlled), and if speed is somehow controlled, the fan flow rate associated with that speed can be measured and the flow coefficient can be determined.

Noise and Interference

Just how extraneous variables affect measured data can be delineated into noise and interference. *Noise* is a random variation of the value of the measured signal as a consequence of the variation of the extraneous variables. Noise increases data scatter. *Interference* imposes undesirable deterministic trends on the measured value. Any uncontrolled influence that causes the signal or test outcome to behave in a manner different from its true behavior is interference.

A common interference in electrical instruments comes from an AC power source and is seen as a sinusoidal wave superimposed onto the measured signal path. Hum and acoustic feedback in public address and audio systems are ready examples of interference effects that are superimposed onto a desirable signal. Sometimes the interference is obvious. But if the period of the interference is longer than the period over which the measurement is made, the false trend may go unnoticed. So we want either to control the source of interference or to break up its trend.

Consider the effects of noise and interference on the signal, $y(t) = 2 + \sin 2\pi t$. As shown in Figure 1.7, noise adds to the scatter of the signal. Through statistical techniques and other means, we can sift through the noise to get at the desirable signal information. But interference imposes a trend onto the signal. The measurement plan should be devised to break up such trends so that they appear as random variations in the data set. Although this will increase the scatter in the measured values of a data set, noise can be handled by statistics. It is far more important to eliminate false trends in the data set.

With this discussion in mind, recall the boiling point example earlier. Barometric pressure caused interference in each individual test. The barometric pressure did not change over the conduct of any one test. But we could discern the effect only because we showed the results of several tests taken over a period for which the value of this uncontrolled variable did change. This is a form of randomization in that as barometric pressure changed between tests, its effect was entered into the data set. Randomization methods are available that can be easily incorporated into the measurement

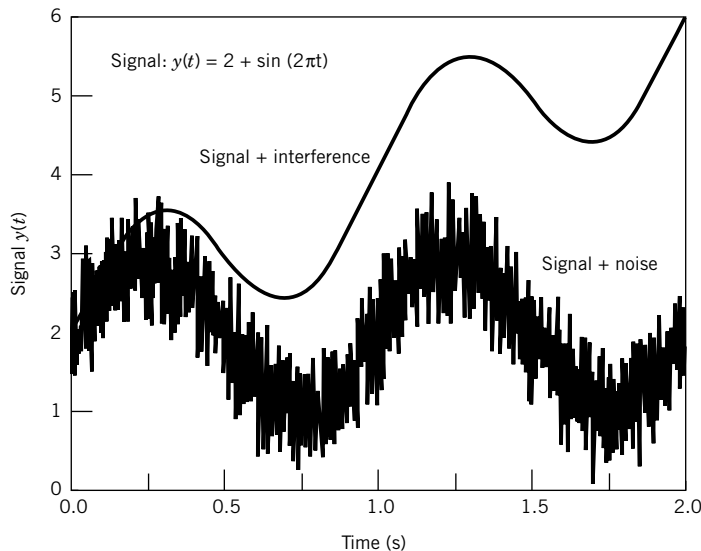


Figure 1.7 Effects of noise and interference superimposed on the signal $y(t) = 2 + \sin 2\pi t$.

plan and will minimize or eliminate interference trends. Several methods are discussed in the paragraphs that follow.

Random Tests

Recall our car fuel-usage example in which the question is: “What fuel usage should I expect from this car?” Let y be the fuel use, which depends on x_a , fuel volume consumption, and x_b , distance traveled. We determine y by varying these two variables (that is, we drive the car). But the test result can be affected by discrete extraneous variables such as the route, driver, weather, and road conditions. For example, driving only on interstate highways would impose a false (untypical) trend on our intended average fuel estimate, so we could drive on different types of roads to break up this trend. This approach introduces a random test strategy.

In general, consider the situation in which the dependent variable, y , is a function of several independent variables, $x_a, x_b, \dots$. However, the measurement of y can also be influenced by several extraneous variables, z_j , where $j = 1, 2, \dots$, such that $y = f(x_a, x_b, \dots; z_j)$. To find the dependence of y on the independent variables, they are varied in a controlled manner. Although the influence of the z_j variables on these tests cannot be eliminated, the possibility of their introducing a false trend on y can be minimized by a proper test strategy. Randomization is one such strategy.

Randomization

We define a *random test* by a measurement matrix that sets a random order to the change in the value of the independent variable applied. The effect of the random order on the results of the test is termed *randomization*. Trends normally introduced by the coupling of a relatively slow and uncontrolled variation in the extraneous variables with a sequential application in values of the independent variable applied will be broken up. This type of plan is effective for the local control of extraneous variables that change in a continuous manner. Consider Examples 1.1 and 1.2.

Discrete extraneous variables are treated a little differently. The use of different instruments, different test operators, and different test operating conditions are examples of discrete extraneous variables that can affect the outcome of a measurement. Randomizing a test matrix to minimize discrete influences can be done efficiently through the use of experimental design using random blocks. A block consists of a data set of the measured variable in which the controlled variable is varied but the extraneous variable is fixed. The extraneous variable is varied between blocks. This enables some amount of local control over the discrete extraneous variable. In the fuel-usage example, we might consider several blocks, each comprised of a different driver (extraneous variable) driving similar routes, and averaging the results. In the example of Figure 1.6, if we cannot control the barometric pressure in the test, then the imposed strategy of using several tests (blocks) under different values of barometric pressure breaks up the interference effect found in a single test. Many strategies for randomized blocks exist, as do advanced statistical methods for data analysis (5–8). In any event, a random test is useful to assess the influence of an uncontrolled variable. Consider Examples 1.3 and 1.4.

Example 1.1

In the pressure calibration system shown in Figure 1.8, a sensor–transducer is exposed to a known pressure, p . The transducer, powered by an external supply, converts the sensed signal into a voltage that is measured by a voltmeter. The measurement approach is to control the applied

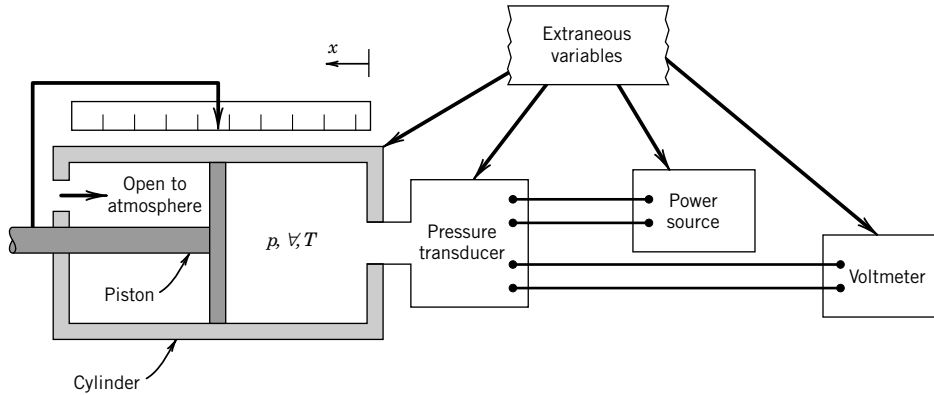


Figure 1.8 Pressure calibration system.

pressure by the measured displacement of a piston that is used to compress a gas contained within the piston-cylinder chamber. The gas chosen closely obeys the ideal gas law. Hence, piston displacement, x , which sets the chamber volume, $V = (x \times \text{area})$, is easily related to chamber pressure. Identify the independent and dependent variables in the calibration and possible extraneous variables.

KNOWN Pressure calibration system of Figure 1.8.

FIND Independent, dependent, and extraneous variables.

SOLUTION The control parameter for this problem can be formed from the ideal gas law: $pV/T = \text{constant}$, where T is the gas temperature. An independent variable in the calibration is the piston displacement that sets the volume. This variable can be controlled by locking the piston into position. From the ideal gas law, gas pressure will also be dependent on temperature, and therefore temperature is also an independent variable. However, T and V are not in themselves independent according to the control parameter. Since volume is to be varied through variation of piston displacement, T and V can be controlled provided a mechanism is incorporated into the scheme to maintain a constant gas temperature within the chamber. This will also maintain chamber area constant, a relevant factor in controlling the volume. In that way, the applied variations in V will be the only effect on pressure, as desired. The dependent variable is the chamber gas pressure. The pressure sensor is exposed to the chamber gas pressure and, hence, this is the pressure it senses. Examples of likely extraneous variables would include noise effects due to the room temperature, z_1 , and line voltage variations, z_2 , which would affect the excitation voltage from the power supply and the performance of the voltmeter. Connecting wires between devices will act as an antenna and possibly will introduce interference, z_3 , superimposed onto the electrical signal, an effect that can be reduced by proper electrical shielding. This list is not exhaustive but illustrative. Hence,

$$p = f(V, T; z_1, z_2, z_3), \text{ where } V = f_1(x, T).$$

COMMENT Even though we might try to keep the gas temperature constant, even slight variations in the gas temperature would affect the volume and pressure and, hence, will act as an additional extraneous variable!

Example 1.2

Develop a test matrix that will minimize the interference effects of any extraneous variables in Example 1.1.

KNOWN $p = f(\forall, T; z_1, z_2, z_3)$, where $\forall = f_1(x; T)$. Control variable $\forall$ is changed. Dependent variable p is measured.

FIND Randomize possible effects of extraneous variables.

SOLUTION Part of our test strategy is to vary volume, control gas temperature, and measure pressure. An important feature of all test plans is a strategy that minimizes the superposition of false trends onto the data set by the extraneous variables. Since z_1 , z_2 , and z_3 and any inability to hold the gas temperature constant are continuous extraneous variables, their influence on p can be randomized by a random test. This entails shuffling the order by which $\forall$ is applied. Say that we pick six values of volume, $\forall_1$, $\forall_2$, $\forall_3$, $\forall_4$, $\forall_5$, and $\forall_6$, where the subscripts correspond to an increasing sequential order of the respective values of volume. Any random order will do fine. One possibility, found by using the random function features of a hand-held calculator, is

$$\forall_2 \forall_5 \forall_1 \forall_4 \forall_6 \forall_3$$

If we perform our measurements in a random order, interference trends will be broken up.

Example 1.3

The manufacture of a particular composite material requires mixing a percentage by weight of binder with resin to produce a gel. The gel is used to impregnate a fiber to produce the composite material in a manual process called the lay-up. The strength, σ , of the finished material depends on the percent binder in the gel. However, the strength may also be lay-up operator dependent. Formulate a test matrix by which the strength to percent binder–gel ratio under production conditions can be established.

KNOWN $\sigma = f(\text{binder}; \text{operator})$

ASSUMPTION Strength is affected only by binder and operator.

FIND Test matrix to randomize effects of operator.

SOLUTION The dependent variable, σ , is to be tested against the independent variable, percent binder–gel ratio. The operator is an extraneous variable in actual production. As a simple test, we could test the relationship between three binder–gel ratios, A , B , and C , and measure strength. We could also choose three typical operators (z_1 , z_2 , and z_3) to produce N separate composite test samples for each of the three binder–gel ratios. This gives the three-block test pattern:

<i>Block</i>				
1	z_1 :	A	B	C
2	z_2 :	A	B	C
3	z_3 :	A	B	C

In the analysis of the test, all of these data can be combined. The results of each block will include each operator's influence as a variation. We can assume that the order used within each block is unimportant. But if only the data from one operator are considered, the results may show a trend consistent with the lay-up technique of that operator. The test matrix above will randomize the influence of any one operator on the strength test results by introducing the influence of several operators.

Example 1.4

Suppose following lay-up, the composite material of Example 1.3 is allowed to cure at a controlled but elevated temperature. We wish to develop a relationship between the binder–gel ratio and the cure temperature and strength. Develop a suitable test matrix.

KNOWN $\sigma = f(\text{binder, temperature, operator})$

ASSUMPTION Strength is affected only by binder, temperature, and operator.

FIND Test matrix to randomize effect of operator.

SOLUTION We develop a simple matrix to test for the dependence of composite strength on the independent variables of binder–gel ratio and cure temperature. We could proceed as in Example 1.3 and set up three randomized blocks for ratio and three for temperature for a total of 18 separate tests. Suppose instead we choose three temperatures, T_1 , T_2 , and T_3 , along with three binder–gel ratios, A , B , and C , and three operators, z_1 , z_2 , and z_3 , and set up a 3×3 test matrix representing a single randomized block. If we organize the block such that no operator runs the same test combination more than once, we randomize the influence of any one operator on a particular binder–gel ratio, temperature test.

	z_1	z_2	z_3
A	T_1	T_2	T_3
B	T_2	T_3	T_1
C	T_3	T_1	T_2

COMMENT The suggested test matrix not only randomizes the extraneous variable, it has reduced the number of tests by one-half over the direct use of three blocks for ratio and for temperature. However, either approach is fine. The above matrix is referred to as a Latin square (5–8).

If we wanted to include our ability to control the independent variables in the test data variations, we could duplicate the Latin-square test several times to build up a significant database.

Replication and Repetition

In general, the estimated value of a measured variable improves with the number of measurements. For example, a bearing manufacturer would obtain a better estimate of the mean diameter and the variation in the diameters of a batch of bearings by measuring many bearings rather than just a few.

Repeated measurements made during any single test run or on a single batch are called *repetitions*. Repetition helps to quantify the variation in a measured variable as it occurs during any one test or batch while the operating conditions are held under nominal control. However, repetition will not permit an assessment of how precisely the operating conditions can be set.

If the bearing manufacturer was interested in how closely bearing mean diameter was controlled in day-in and day-out operations with a particular machine or test operator, duplicate tests run on different days would be needed. An independent duplication of a set of measurements using similar operating conditions is referred to as a *replication*. Replication allows for quantifying the variation in a measured variable as it occurs between different tests, each having the same nominal values of operating conditions.

Finally, if the bearing manufacturer were interested in how closely bearing mean diameter was controlled when using different machines or different machine operators, duplicate tests using these different configurations holds the answer. Here, replication provides a means to randomize the interference effects of the different bearing machines or operators.

Replication allows us to assess the control of setting the operating conditions, that is, the ability to reset the conditions to some desired value. Ultimately, replication provides the means to estimate control over the procedure used.

Example 1.5

Consider a room furnace thermostat. Set to some temperature, we can make repeated measurements (repetition) of room temperature and come to a conclusion about the average value and the variation in room temperature at that particular thermostat setting. Repetition allows us to estimate the variation in this measured variable. This repetition permits an assessment of how well we can maintain (control) the operating condition.

Now suppose we change the set temperature to some arbitrary value but sometime later return it to the original setting and duplicate the measurements. The two sets of test data are replications of each other. We might find that the average temperature in the second test differs from the first. The different averages suggest something about our ability to set and control the temperature in the room. Replication permits the assessment of how well we can duplicate a set of conditions.

Concomitant Methods

Is my test working? What value of result should I expect? To help answer these, a good strategy is to incorporate *concomitant methods* in a measurement plan. The goal is to obtain two or more estimates for the result, each based on a different method, which can be compared as a check for agreement. This may affect the experimental design in that additional variables may need to be measured. Or the different method could be an analysis that estimates an expected value of the measurement. For example, suppose we want to establish the volume of a cylindrical rod of known material. We could simply measure the diameter and length of the rod to compute this. Alternatively, we could measure the weight of the rod and compute volume based on the specific weight of the material. The second method complements the first and provides an important check on the adequacy of the first estimate.

1.4 CALIBRATION

A *calibration* applies a known input value to a measurement system for the purpose of observing the system output value. It establishes the relationship between the input and output values. The known value used for the calibration is called the *standard*.

Static Calibration

The most common type of calibration is known as a static calibration. In this procedure, a known value is input to the system under calibration and the system output is recorded. The term “static” implies that the values of the variables involved remain constant; that is, they do not vary with time or space. In static calibrations, only the magnitudes of the known input and the measured output are important.

By applying a range of known input values and by observing the system output values, a direct calibration curve can be developed for the measurement system. On such a curve the input, x , is plotted on the abscissa against the measured output, y , on the ordinate, such as indicated in Figure 1.9. In a calibration the input value is usually a controlled independent variable, while the measured output value is the dependent variable of the calibration.

The static calibration curve describes the static input–output relationship for a measurement system and forms the logic by which the indicated output can be interpreted during an actual measurement. For example, the calibration curve is the basis for fixing the output display scale on a measurement system, such as that of Figure 1.4. Alternatively, a calibration curve can be used as part of developing a functional relationship, an equation known as a correlation, between input and output. A correlation will have the form $y = f(x)$ and is determined by applying physical reasoning and curve fitting techniques to the calibration curve. The correlation can then be used in later measurements to ascertain the unknown input value based on the output value, the value indicated by the measurement system.

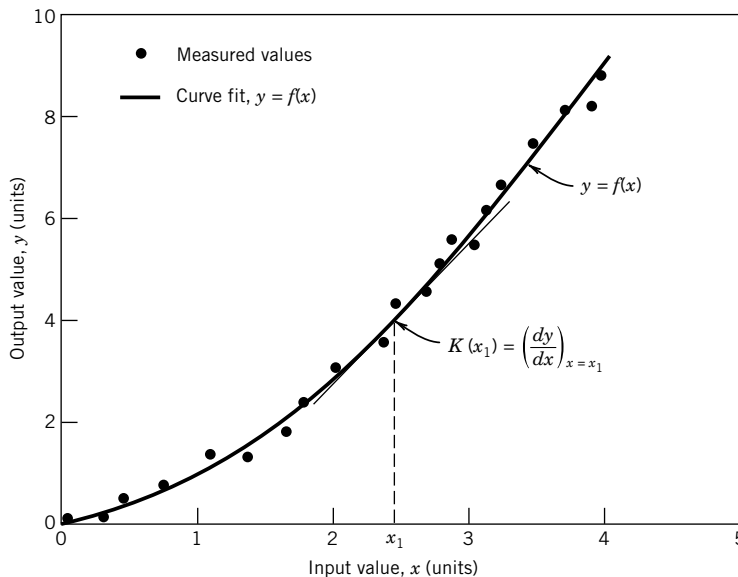


Figure 1.9 Representative static calibration curve.

Dynamic Calibration

When the variables of interest are time (or space) dependent and such varying information is sought, we need dynamic information. In a broad sense, dynamic variables are time (or space) dependent in both their magnitude and frequency content. A *dynamic calibration* determines the relationship between an input of known dynamic behavior and the measurement system output. Usually, such calibrations involve applying either a sinusoidal signal or a step change as the known input signal. The dynamic nature of signals and measurement systems is explored fully in Chapter 3.

Static Sensitivity

The slope of a static calibration curve provides the static sensitivity⁴ of the measurement system. As depicted graphically in the calibration curve of Figure 1.9, the static sensitivity, K , at any particular static input value, say x_1 , is evaluated by

$$K = K(x_1) = \left(\frac{dy}{dx} \right)_{x=x_1} \quad (1.1)$$

where K is a function of x . The static sensitivity is a measure relating the change in the indicated output associated with a given change in a static input.

Range

A calibration applies known inputs ranging from the minimum to the maximum values for which the measurement system is to be used. These limits define the operating *range* of the system. The input operating range is defined as extending from $x_{\min}$ to $x_{\max}$. The input operating range may be expressed in terms of the difference of the limits as

$$r_i = x_{\max} - x_{\min} \quad (1.2)$$

This is equivalent to specifying the output operating range from $y_{\min}$ to $y_{\max}$. The output span or full-scale operating range (FSO) is expressed as

$$r_o = y_{\max} - y_{\min} \quad (1.3)$$

It is important to avoid extrapolation beyond the range of known calibration during measurement since the behavior of the measurement system is uncharted in these regions. As such, the range of calibration should be carefully selected.

Resolution

The *resolution* represents the smallest increment in the measured value that can be discerned. In terms of a measurement system, it is quantified by the smallest scale increment or least count (least significant digit) of the output readout indicator.

⁴ Some texts refer to this as the *static gain*.

Accuracy and Error

The exact value of a variable is called the *true value*. The value of the variables as indicated by a measurement system is called the *measured value*. The *accuracy* of a measurement refers to the closeness of agreement between the measured value and the true value. But the true value is rarely known exactly, and various influences, called errors, have an effect on both of these values. So the concept of the accuracy of a measurement is a qualitative one.

An appropriate approach to stating this closeness of agreement is to identify the measurement errors and to quantify them by the value of their associated uncertainties, where an uncertainty is the estimated range of value of an error. We define an *error*, e , as the difference between the measured value and the true value, that is

$$e = \text{Measured value} - \text{True value} \quad (1.4)$$

While the true value is rarely known exactly, Equation 1.4 serves as a reference definition. Errors exist and they have a magnitude as given by Equation 1.4. The concept is something we discuss next and then develop extensively in Chapter 5.

Often an estimate for the value of error is based on a reference value used during the instrument's calibration as a surrogate for the true value. A relative error based on this reference value is estimated by

$$A = \frac{|e|}{\text{Reference value}} \times 100 \quad (1.5)$$

A few vendors may still refer to this term as the “relative accuracy.”

A special form of a calibration curve is the *deviation plot*, such as shown in Figure 1.10. Such a curve plots the error or deviation between a reference or expected value, y' , and the measured value, y , versus the measured value. Deviation curves are extremely useful when the differences between the reference and the measured value are too small to suggest possible trends on direct calibration plots. As an example, a deviation plot of the calibration of a temperature-sensing thermocouple is

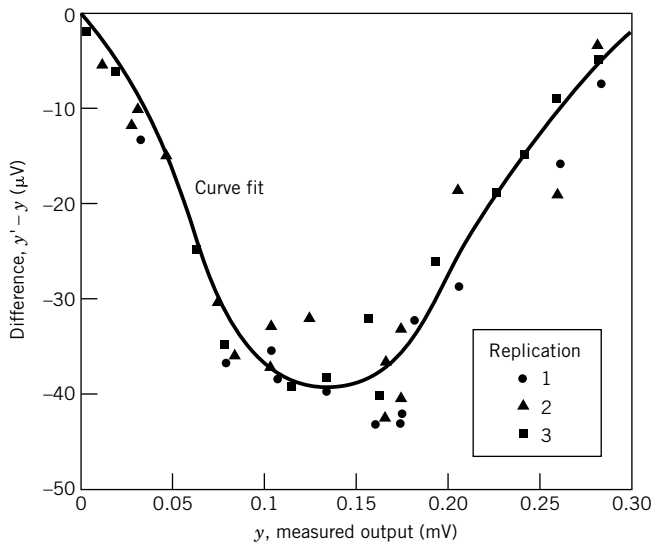


Figure 1.10 Calibration curve in the form of a deviation plot for a temperature sensor.

given in Figure 1.10. The voltage output during calibration is compared with the expected values obtained from reference tables. The trend of the errors can be correlated, as shown by the curve fit. We see the maximum errors are two orders of magnitude smaller than the measured values, but we also see that the errors are smallest at either limit of the measuring range. We also see that the error varies in magnitude. The range over which it varies is the uncertainty in the measurement.

Random and Systematic Errors and Uncertainty

Errors are effects that cause a measured value to differ from its true value. *Random error* causes a random variation in measured values found during repeated measurements of a variable. *Systematic error* causes an offset between the mean value of the data set and its true value. Both random and systematic errors affect a system's accuracy.

The concept of accuracy and the effects of systematic and random errors in instruments and measurement systems can be illustrated by the throw of darts. Consider the dart boards in Figure 1.11 where the goal will be to throw the darts into the bull's-eye. For this analogy, the bull's-eye can represent the true value and each throw can represent a measured value. In Figure 1.11a, the thrower displays good repeatability (i.e., a small effect from random error) in that each throw repeatedly hits the same spot on the board, but the thrower is not accurate in that the dart misses the bull's-eye each time. We see that a small amount of random error is not a complete measure of the accuracy of this thrower. The error in each throw can be computed from the distance between the bull's-eye and each dart. The average value of the error gives an estimate of the systematic error in the throws. This thrower has an offset to the left of the target. If the effect of this systematic error could be reduced, then this thrower's accuracy would improve. In Figure 1.11b, the thrower displays a high accuracy, hitting the bull's-eye on each throw. Both scatter and offset are near zero. High accuracy must imply a small influence of both the random and systematic errors as shown. In Figure 1.11c, the thrower does not show good accuracy, with errant throws scattered around the board. Each throw contains a different amount of error. While an estimate of the systematic error is the average of the errors in the throws, the estimate of the random error is related to the varying amount of error in the throws, a value that can be estimated using statistical methods. The estimates in the random and systematic errors of the thrower can be computed using the statistical methods that are discussed in Chapter 4 or the methods of comparison discussed in Chapter 5.

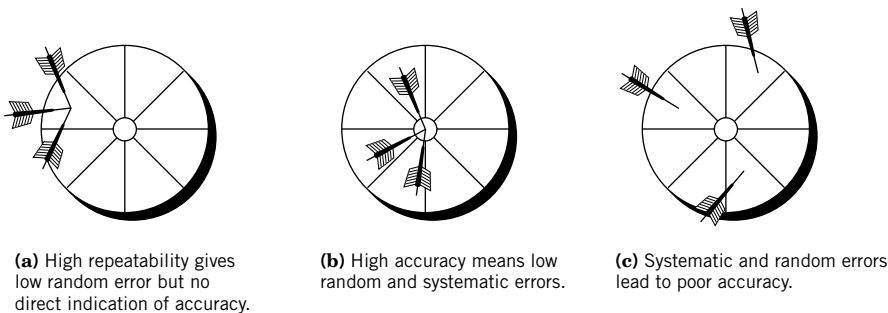


Figure 1.11 Throws of a dart: illustration of random and systematic errors and accuracy.

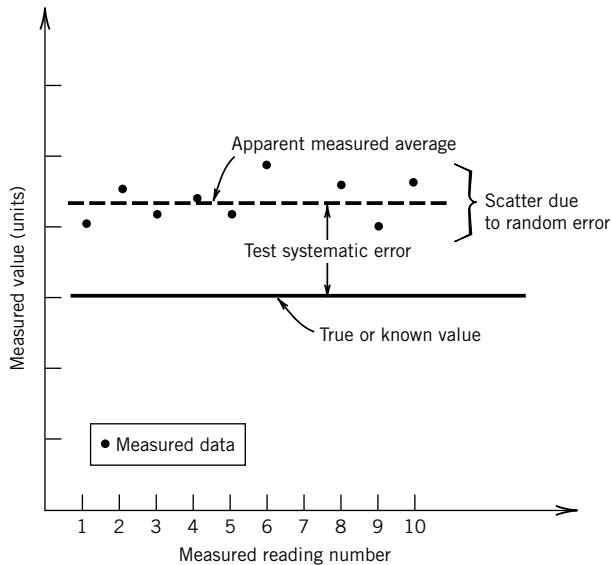


Figure 1.12 Effects of random and systematic errors on calibration readings.

Suppose we used a measurement system to measure a variable whose value was kept constant and known almost exactly, as in a calibration. For example, 10 independent measurements are made with the results as shown in Figure 1.12. The variations in the measurements, the observed scatter in the data, would be related to the random error associated with the measurement of the variable. This scatter is mainly due to (1) the measurement system and the measurement method, and (2) any uncontrolled variations in the variable. However, the offset between the apparent average of the readings and the true value would provide a measure of the systematic error to be expected from this measurement system.

Uncertainty

The *uncertainty* is a numerical estimate of the possible range of the error in a measurement. In any measurement, the error is not known exactly since the true value is rarely known exactly. But based on available information, the operator might feel confident that the error is within certain bounds, a plus or minus range of the indicated reading. This is the assigned uncertainty. Uncertainty is brought about by all of the errors that are present in the measurement system—its calibration, the data set statistics, and the measurement technique. Individual errors are properties of the instruments, the test method, the analysis, and the measurement system. Uncertainty is a property of the test result. In Figure 1.12, we see that we might assign an estimate to the random error, that is, the random uncertainty, based on the data scatter. The systematic uncertainty might be based on a comparison against a concomitant method. A method of estimating the overall uncertainty in the test result is treated in detail in Chapter 5.

The uncertainty values assigned to an instrument or measurement system specification are usually the result of several interacting random and systematic errors inherent to the measurement system, the calibration procedure, and the standard used to provide the known value. An example of the errors affecting an instrument is given for a typical pressure transducer in Table 1.1. The value assigned to each error is the uncertainty.

Table 1.1 Manufacturer's Specifications: Typical Pressure Transducer

Operation	
Input range	0–1000 cm H ₂ O
Excitation	±15 V DC
Output range	0–5 V
Performance	
Linearity error	±0.5% FSO
Hysteresis error	Less than ±0.15% FSO
Sensitivity error	±0.25% of reading
Thermal sensitivity error	±0.02%/°C of reading
Thermal zero drift	±0.02%/°C FSO
Temperature range	0–50 °C

FSO, full-scale operating range.

Sequential Test

A *sequential test* applies a sequential variation in the input value over the desired input range. This may be accomplished by increasing the input value (upscale direction) or by decreasing the input value (downscale direction) over the full input range.

Hysteresis

The sequential test is an effective diagnostic technique for identifying and quantifying hysteresis error in a measurement system. *Hysteresis error* refers to differences between an upscale sequential test and a downscale sequential test. The hysteresis error of the system is estimated by its uncertainty $u_h = (y)_{\text{upscale}} - (y)_{\text{downscale}}$. The effect of hysteresis in a sequential test calibration curve is illustrated in Figure 1.13a. Hysteresis is usually specified for a measurement system in terms of the maximum hysteresis error as a percentage of full-scale output range, r_o ,

$$\%u_{h_{\max}} = \frac{u_{h_{\max}}}{r_o} \times 100 \quad (1.6)$$

such as the value indicated in Table 1.1. Hysteresis occurs when the output of a measurement system is dependent on the previous value indicated by the system. Such dependencies can be brought about through some realistic system limitations such as friction or viscous damping in moving parts or residual charge in electrical components. Some hysteresis is normal for any system and affects the repeatability of the system.

Random Test

A *random test* applies a random order in the values of a known input over the intended calibration range. The random application of input tends to reduce the impact of interference. It breaks up hysteresis effects and observation errors. It ensures that each application of input value is independent of the previous. As such, it reduces calibration systematic error, converting it to random error. Generally, such a random variation in input value will more closely simulate the actual measurement situation.

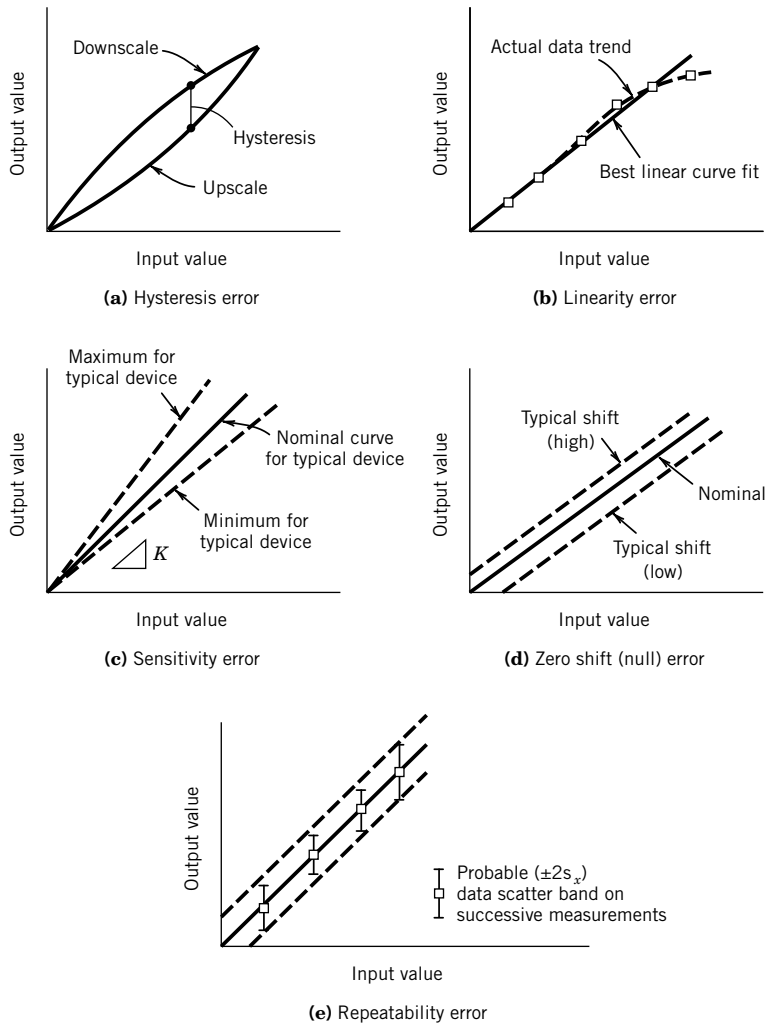


Figure 1.13 Examples of some common elements of instrument error.

- (a) Hysteresis error.
- (b) Linearity error.
- (c) Sensitivity error.
- (d) Zero shift (null) error.
- (e) Repeatability error.

A random test provides an important diagnostic for the delineation of several measurement system performance characteristics based on a set of random calibration test data. In particular, linearity error, sensitivity error, zero error, and instrument repeatability error, as illustrated in Figure 1.13b–e, can be quantified from a static random test calibration.

Linearity Error

Many instruments are designed to achieve a linear relationship between the applied static input and indicated output values. Such a linear static calibration curve would have the general form

$$y_L(x) = a_0 + a_1x \quad (1.7)$$

where the curve fit $y_L(x)$ provides a predicted output value based on a linear relation between x and y . However, in real systems, truly linear behavior is only approximately achieved. As a result,

measurement device specifications usually provide a statement as to the expected linearity of the static calibration curve for the device. The relationship between $y_L(x)$ and measured value $y(x)$ is a measure of the nonlinear behavior of a system:

$$u_L(x) = y(x) - y_L(x) \quad (1.8)$$

where $u_L(x)$ is a measure of the *linearity error* that arises in describing the actual system behavior by Equation 1.7. Such behavior is illustrated in Figure 1.13b in which a linear curve has been fit through a calibration data set. For a measurement system that is essentially linear in behavior, the extent of possible nonlinearity in a measurement device is often specified in terms of the maximum expected linearity error as a percentage of full-scale output range, r_o ,

$$\%u_{L_{\max}} = \frac{u_{L_{\max}}}{r_o} \times 100 \quad (1.9)$$

This is how the linearity error for the pressure transducer in Table 1.1 was estimated. Statistical methods of quantifying data scatter about a line or curve fit are discussed in Chapter 4.

Sensitivity and Zero Errors

The scatter in the data measured during a calibration affects the precision in predicting the slope of the calibration curve. As shown for the linear calibration curve in Figure 1.13c, in which the zero intercept is fixed, the scatter in the data about the curve fit are random errors. The *sensitivity error*, u_K , is a statistical measure of the random error in the estimate of the slope of the calibration curve (we discuss the statistical estimate further in Chapter 4). The static sensitivity of a device is also temperature dependent, and this is often specified. In Table 1.1, the sensitivity error reflects calibration results at a constant reference ambient temperature, whereas the thermal sensitivity error was found by calibration at different temperatures.

If the zero intercept is not fixed but the sensitivity is constant, then a drift in the zero intercept introduces a vertical shift of the calibration curve, as shown in Figure 1.13d. This shift is known as the *zero error* with uncertainty, u_z . Zero error can usually be reduced by periodically adjusting the output from the measurement system under a zero input condition. However, some random variation in the zero intercept is common, particularly with electronic and digital equipment subjected to temperature variations (e.g., thermal zero drift in Table 1.1).

Instrument Repeatability

The ability of a measurement system to indicate the same value on repeated but independent application of the same input provides a measure of the instrument *repeatability*. Specific claims of repeatability are based on multiple calibration tests (replication) performed within a given lab on the particular unit. Repeatability, as shown in Figure 1.13e, is based on a statistical measure (developed in Chapter 4) called the standard deviation, s_x , a measure of the variation in the output for a given input. The value claimed is usually in terms of the maximum expected error as a percentage of full-scale output range:

$$\%u_{R_{\max}} = \frac{2s_x}{r_o} \times 100 \quad (1.10)$$

The instrument repeatability reflects only the variations found under controlled calibration conditions.

Reproducibility

The term “reproducibility,” when reported in instrument specifications, refers to the closeness of agreement in results obtained from duplicate tests carried out under similar conditions of measurement. As with repeatability, the uncertainty is based on statistical measures. Manufacturer claims of instrument reproducibility must be based on multiple tests (replication) performed in different labs on a single unit or model of instrument.

Instrument Precision

The term “instrument precision,” when reported in instrument specifications, refers to a random uncertainty based on the results of separate repeatability tests. Manufacturer claims of instrument precision must be based on multiple tests (replication) performed on different units of the same manufacture, either performed in the same lab (same-lab precision) or, preferably, performed in different labs (between-lab precision).

Overall Instrument Error and Instrument Uncertainty

An estimate of the *overall instrument error* is made by combining the estimates of all known errors into a term called the *instrument uncertainty*. The estimate is computed from the square root of the sum of the squares of all known uncertainty values. For M known errors, the overall instrument uncertainty, u_c , is estimated by

$$u_c = [u_1^2 + u_2^2 + \cdots + u_M^2]^{1/2} \quad (1.11)$$

For example, for an instrument having known hysteresis, linearity, and sensitivity errors, the instrument uncertainty is estimated by

$$u_c = [u_h^2 + u_L^2 + u_K^2]^{1/2} \quad (1.12)$$

1.5 STANDARDS

When a measurement system is calibrated, its indicated value is compared directly with a reference value. This reference value forms the basis of the comparison and is known as the *standard*. This standard may be based on the output from a piece of equipment, from an object having a well-defined physical attribute to be used as a comparison, or from a well-accepted technique known to produce a reliable value. Let us explore how certain standards come to be and how these standards are the foundation of all measurements.

Primary Standards

A *dimension* defines a physical variable that is used to describe some aspect of a physical system. A *unit* defines a quantitative measure of a dimension. For example, mass, length, and time describe base dimensions with which we associate the units of kilogram, meter, and second. A *primary standard* defines the value of a unit. It provides the means to describe the unit with a unique number that can be understood throughout the world. The primary standard, then, assigns a unique value to a unit by definition! As such it must define the unit exactly. In 1960, the General Conference on Weights and Measures (CGPM), the international agency responsible for maintaining exact uniform standards of measurements, formally adopted the International System of Units (SI) as the

international standard of units. The system has been adopted worldwide. Other unit systems are commonly used in the consumer market and so deserve mention. These other unit systems are not standards and are treated as conversions from SI. Examples of these include the inch-pound (I-P) unit system found in the United States and the gravitational mks (meter-kilogram-second or metric) unit system common to much of the world.

Primary standards are necessary because the value assigned to a unit is actually arbitrary. For example, over 4500 years ago the Egyptian cubit was used as a standard of length and based on the length from outstretched fingertips to the elbow. It was later codified with a master of marble, a stick about 52 cm in length, on which scratches were etched to define subunits of length. This standard served well for centuries!

So whether today's standard unit of length, the meter, is the length of a king's forearm or the distance light travels in a fraction of a second really only depends on how we want to define it. To avoid confusion, units are defined by international agreement through the use of primary standards. Once agreed upon, a primary standard forms the exact definition of the unit until it is changed by some later agreement. Important features sought in any standard should include global availability, continued reliability, and stability with minimal sensitivity to external environmental sources. Next we examine some basic dimensions and the primary standards that form the definition of the units that describe them (9).

Base Dimensions and Their Units

Mass

The dimension of mass is defined by the kilogram. Originally, the unit of the kilogram was defined by the mass of one liter of water at room temperature. But today an equivalent yet more consistent definition defines the kilogram exactly as the mass of a particular platinum-iridium cylindrical bar that is maintained under very specific conditions at the International Bureau of Weights and Measures located in Sevres, France. This particular bar (consisting of 90% platinum and 10% iridium by mass) forms the primary standard for the kilogram. It remains today as the only basic unit still defined in terms of a material object.

In the United States, the I-P unit system (also referred to as the U.S. customary units) remains widely used. In the I-P system, *mass* is defined by the pound-mass, lb_m , which is derived directly and exactly from the definition of the kilogram:

$$1 \text{ lb}_m = 0.4535924 \text{ kg} \quad (1.13)$$

Equivalent standards for the kilogram and other standards units are maintained by the U.S. National Institute of Standards and Technology (NIST) in Gaithersburg, Maryland, and other national labs around the globe. NIST claims that their mass standard is accurate to an uncertainty of within 1 mg in 27,200 kg.

Time and Frequency

The dimension of time is defined by the unit of a second. One second (s) is defined as the time elapsed during 9,192,631,770 periods of the radiation emitted between two excitation levels of the fundamental state of cesium-133 (10). Despite this seemingly unusual definition, this primary standard can be reliably reproduced at suitably equipped laboratories throughout the world to an uncertainty of within two parts in 10 trillion.

The Bureau International de l'Heure (BIH) in Paris maintains the primary standard for clock time. Periodically, adjustments to clocks around the world are made relative to the BIH clock so as to keep time synchronous.

The standard for cyclical frequency is derived from the time standard. The standard unit is the hertz (1 Hz = 1 cycle/s). The cyclical frequency is related to the circular frequency (radians/s) by

$$1 \text{ Hz} = \frac{2\pi \text{ rad}}{1 \text{ s}} \quad (1.14)$$

Both time and frequency standard signals are broadcast worldwide over designated radio stations for use in navigation and as a source of a standard for these dimensions.

Length

The meter is the standard unit for length. New primary standards are established when our ability to determine the new standard becomes more accurate (i.e., lower uncertainty) than the existing standard. In 1982, a new primary standard was adopted by the CGPM to define the unit of a meter. One meter (m) is now defined exactly as the length traveled by light in $1/299,792,458$ of a second, a number derived from the velocity of light in a vacuum (defined as $299,792,458 \text{ m/s}$).

The I-P system unit of the inch and the related unit of the foot are derived exactly from the meter.

$$\begin{aligned} 1 \text{ ft} &= 0.3048 \text{ m} \\ 1 \text{ in.} &= 0.0254 \text{ m} \end{aligned} \quad (1.15)$$

Temperature

The kelvin, K, is the SI unit of thermodynamic temperature and is the fraction $1/273.15$ of the thermodynamic temperature of the triple point of water. This temperature scale was devised by William Thomson, Lord Kelvin (1824–1907), and forms the basis for the absolute practical temperature scale in common use. This scale is based on polynomial interpolation between the equilibrium phase change points of a number of common pure substances from the triple point of equilibrium hydrogen (13.81 K) to the freezing point of pure gold (1337.58 K). Above 1337.58 K, the scale is based on Planck's law of radiant emissions. The details of the standard have been modified over the years but are governed by the International Temperature Scale–1990 (11).

The I-P unit system uses the absolute scale of Rankine ($^{\circ}\text{R}$). This and the common scales of Celsius ($^{\circ}\text{C}$), used in the metric system, and Fahrenheit ($^{\circ}\text{F}$) are related to the Kelvin scale by the following:

$$\begin{aligned} (^{\circ}\text{C}) &= (\text{K}) - 273.15 \\ (^{\circ}\text{F}) &= (^{\circ}\text{R}) - 459.67 \\ (^{\circ}\text{F}) &= 1.8 \times (^{\circ}\text{C}) + 32.0 \end{aligned} \quad (1.16)$$

Current

The SI unit for current is the ampere. One ampere (A) is defined as that constant current which, if maintained in two straight parallel conductors of infinite length and of negligible circular cross

section and placed 1 m apart in vacuum, would produce a force equal to 2×10^{-7} newtons per meter of length between these conductors.

Measure of Substance

The unit of quantity of a substance is defined by the mole. One mole (mol) is the amount of substance of a system that contains as many elementary entities as there are atoms in 0.012 kilogram of carbon 12.

Luminous Intensity

The intensity of light is defined by the candela. One candela (cd) is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 5.40×10^{14} hertz and that has a radiant intensity in that direction of 1/683 watt per steradian.

Derived Units

Other dimensions and their associated units are defined in terms of and derived from the base dimensions and units (9,12).

Force

From Newton's law, force is proportional to mass times acceleration:

$$\text{Force} = \frac{\text{Mass} \times \text{Acceleration}}{g_c}$$

where g_c is a proportionality constant used to maintain consistency in units.

Force is defined by a derived unit called the newton (N), which is derived from the base dimensions of mass, length, and time:

$$1 \text{ N} = 1 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \quad (1.17)$$

So for this system the value of g_c must be $1.0 \text{ kg} \cdot \text{m}/\text{s}^2 \cdot \text{N}$. Note that the resulting expression for Newton's second law does not explicitly require the inclusion of g_c to make units match and so is often ignored.

However, in I-P units, the units of force and mass are related through the definition: One pound-mass (lb_m) exerts a force of 1 pound (lb) in a standard gravitational field. With this definition,

$$1 \text{ lb} = \frac{(1 \text{ lb}_m)(32.174 \text{ ft}/\text{s}^2)}{g_c} \quad (1.18)$$

and g_c must take on the value of $32.174 \text{ lb}_m \cdot \text{ft}/\text{lb} \cdot \text{s}^2$. In I-P units, the pound is a defined quantity and g_c must be derived through Newton's law.

Similarly, in the gravitational mks (metric) system, which uses the kilogram-force (kg_f),

$$1 \text{ kg}_f = \frac{(1 \text{ kg})(9.80665 \text{ m}/\text{s}^2)}{g_c} \quad (1.19)$$

and the value for g_c takes on a value of exactly $9.80665 \text{ kg} \cdot \text{m}/\text{s}^2 \cdot \text{kg}_f$.

Many engineers have some difficulty with using g_c in the non-SI systems. Actually, whenever force and mass appear in the same expression, just remember to relate them using g_c through Newton's law:

$$g_c = \frac{mg}{F} = 1 \frac{\text{kg} \cdot \text{m/s}^2}{\text{N}} = 32.174 \frac{\text{lb}_m \cdot \text{ft/s}^2}{\text{lb}} = 9.80665 \frac{\text{kg} \cdot \text{m/s}^2}{\text{kg}_f} \quad (1.20)$$

Other Derived Dimensions and Units

Energy is defined as force times length and uses the unit of the joule (J), which is derived from base units as

$$1 \text{ J} = 1 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} = 1 \text{ N} \cdot \text{m} \quad (1.21)$$

Power is defined as energy per unit time in terms of the unit of the watt (W), which is derived from base units as:

$$1 \text{ W} = 1 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^3} = 1 \frac{\text{J}}{\text{s}} \quad (1.22)$$

Stress and pressure are defined as force per unit area, where area is length squared, in terms of the pascal (Pa), which is derived from base units as

$$1 \text{ Pa} = 1 \frac{\text{kg}}{\text{m} \cdot \text{s}^2} = 1 \text{ N/m}^2 \quad (1.23)$$

Electrical Dimensions

The units for the dimensions of electrical potential, and resistance, charge, and capacitance are based on the definitions of the absolute volt (V), and ohm (Ω), coulomb (C), and farad (F), respectively. Derived from the ampere, 1 ohm absolute is defined by 0.9995 times the resistance to current flow of a column of mercury that is 1.063 m in length and has a mass of 0.0144521 kg at 273.15 K. The volt is derived from the units for power and current, $1 \text{ V} = 1 \text{ N} \cdot \text{m/s} \cdot \text{A} = 1 \text{ W/A}$. The ohm is derived from the units for electrical potential and current, $1 \Omega = 1 \text{ kg} \cdot \text{m}^2/\text{s}^3 \cdot \text{A}^2 = 1 \text{ V/A}$. The coulomb is derived from the units for current and time, $1 \text{ C} = 1 \text{ A} \cdot \text{s}$. One volt is the difference of potential between two points of an electrical conductor when a current of 1 ampere flowing between those points dissipates a power of 1 watt. The farad (F) is the standard unit for capacitance derived from the units for charge and electric potential, $1 \text{ F} = 1 \text{ C/V}$.

On a practical level, working standards for resistance and capacitance take the form of certified standard resistors and capacitors or resistance boxes and are used as standards for comparison in the calibration of resistance measuring devices. The practical potential standard makes use of a standard cell consisting of a saturated solution of cadmium sulfate. The potential difference of two conductors connected across such a solution is set at 1.0183 V and at 293 K. The standard cell maintains constant electromotive force over very long periods of time, provided that it is not subjected to a current drain exceeding 100 μA for more than a few minutes. The standard cell is typically used as a standard for comparison for voltage measurement devices.

Table 1.2 Dimensions and Units^a

Unit	Dimension	
	SI	I-P
Primary		
Length	meter (m)	inch (in.)
Mass	kilogram (kg)	pound-mass (lb _m)
Time	second (s)	second (s)
Temperature	kelvin (K)	rankine (°R)
Current	ampere (A)	ampere (A)
Substance	mole (mol)	mole (mol)
Light intensity	candela (cd)	candela (cd)
Derived		
Force	newton (N)	pound-force (lb)
Voltage	volt (V)	volt (V)
Resistance	ohm (Ω)	ohm (Ω)
Capacitance	farad (F)	farad (F)
Inductance	henry (H)	henry (H)
Stress, pressure	pascal (Pa)	pound-force/inch ² (psi)
Energy	joule (J)	foot pound-force (ft-lb)
Power	watt (W)	foot pound-force/second (ft-lb/s)

^aSI dimensions and units are the international standards. I-P units are presented for convenience.

A chart for converting between units is included inside the text cover. Table 1.2 lists the basic standard and some derived units used in SI and the corresponding units used in the I-P and gravitational metric systems.

Hierarchy of Standards

The known value applied to a measurement system during calibration becomes the standard on which the calibration is based. So how do we pick this standard, and how good is it? Obviously, actual primary standards are impractical as standards for normal calibration use. But they serve as a reference for exactness. It would not be reasonable to travel to France to calibrate an ordinary laboratory scale using the primary standard for mass (nor would it likely be permitted!). So for practical reasons, there exists a hierarchy of reference and secondary standards used to duplicate the primary standards. Just below the primary standard in terms of absolute accuracy are the national reference standards maintained by designated standards laboratories throughout the world. These provide a reasonable duplication of the primary standard but allow for worldwide access to an extremely accurate standard. Next to these, we develop transfer standards. These are used to calibrate individual laboratory standards that might be used at various facilities within a country. Laboratory standards serve to calibrate working standards. Working standards are used to calibrate everyday devices used in manufacturing and research facilities. In the United States, NIST maintains primary, reference, and secondary standards and recommends standard procedures for the calibration of measurement systems.

Table 1.3 Hierarchy of Standards^a

Primary standard	Maintained as absolute unit standard
Transfer standard	Used to calibrate local standards
Local standard	Used to calibrate working standards
Working standard	Used to calibrate local instruments

^aThere may be additional intermediate standards between each hierarchy level.

Each subsequent level of the hierarchy is derived by calibration against the standard at the previous higher level. Table 1.3 lists an example of such a lineage for standards from a primary or reference standard maintained at a national standards lab down to a working standard used in a typical laboratory or production facility to calibrate everyday working instruments. If the facility does not maintain a local (laboratory or working) standard, then the instruments must be sent off and calibrated elsewhere. In such a case, a standards traceability certificate would be issued for the instrument.

As one moves down through the standards lineage, the degree of exactness by which a standard approximates the primary standard deteriorates. That is, increasing elements of error are introduced into the standard as one moves from one level of hierarchy of standard to the next. As a common example, an institution might maintain its own working standard (for some application) that is used to calibrate the measurement devices found in the individual laboratories throughout the institution. Periodic calibration of the working standard might be against the institution's well-maintained local standard. The local standard would be periodically sent off to be calibrated against the NIST (or appropriate national standards lab) transfer standard (and traceability certificate issued). NIST will periodically calibrate its own transfer standard against its reference or primary standard. This is illustrated for a temperature standard traceability hierarchy in Table 1.4. The uncertainty in the approximation of the known value increases as one moves down the hierarchy. It follows, then, that since the calibration determines the relationship between the input value and the output value, the accuracy of the calibration will depend in part on the accuracy of the standard. But if typical working standards contain errors, how is accuracy ever determined? At best, this closeness of agreement is quantified by the estimates of the known uncertainties in the calibration. And the confidence in that estimate depends on the quality of the standard and the calibration techniques used.

Table 1.4 Example of a Temperature Standard Traceability

Standard		
Level	Method	Uncertainty [$^{\circ}\text{C}$] ^a
Primary	Fixed thermodynamic points	0
Transfer	Platinum resistance thermometer	± 0.005
Working	Platinum resistance thermometer	± 0.05
Local	Thermocouple	± 0.5

^aTypical combined instrument systematic and random uncertainties.

Test Standards and Codes

The term “standard” is also applied in other ways in engineering. *Test standards* refer to well-defined test procedures, technical terminology, methods to construct test specimens or test devices, and/or methods for data reduction. The goal of a test standard is to provide consistency in the conduct and reporting of a certain type of measurement between test facilities. Similarly, *test codes* refer to procedures for the manufacture, installation, calibration, performance specification, and safe operation of equipment.

Diverse examples of test standards and codes are illustrated in readily available documents (13–16) from professional societies, such as the American Society of Mechanical Engineers (ASME), the American Society of Testing and Materials (ASTM), and the International Standards Organization (ISO). For example, ASME Power Test Code 19.5 provides detailed designs and operation procedures for flow meters, while ASTM Test Standard F558-88 provides detailed procedures for evaluating vacuum cleaner cleaning effectiveness and controls the language for product performance claims. Engineering standards and codes are consensus documents agreed on by knowledgeable parties interested in establishing, for example, some common basis for comparing equipment performance between manufacturers. These are not binding legal documents unless specifically adopted and implemented as such by a government agency. Still, they present a convincing argument for best practice.

1.6 PRESENTING DATA

Since we use several plotting formats throughout this text to present data, it is best to introduce these formats here. Data presentation conveys significant information about the relationship between variables. Software is readily available to assist in providing high-quality plots, or plots can be generated manually using graph paper. Several forms of plotting formats are discussed next.

Rectangular Coordinate Format

In rectangular grid format, both the ordinate and the abscissa have uniformly sized divisions providing a linear scale. This is the most common format used for constructing plots and establishing the form of the relationship between the independent and dependent variable.

Semilog Coordinate Format

In a semilog format, one coordinate has a linear scale and one coordinate has a logarithmic scale. Plotting values on a logarithmic scale performs a logarithmic operation on those values, for example, plotting $y = f(x)$ on a logarithmic x -axis is the same as plotting $y = \log f(x)$ on rectangular axes. Logarithmic scales are advantageous when one of the variables spans more than one order of magnitude. In particular, the semilog format may be convenient when the data approximately follow a relationship of the form $y = ae^x$ or $y = a10^x$ as a linear curve will result in each case. A natural logarithmic operation can also be conveyed on a logarithmic scale, as the relation $\ln y = 2.3 \log y$ is just a scaling operation.

Full-Log Coordinate Format

The full-log or log-log format has logarithmic scales for both axes and is equivalent to plotting $\log y$ vs. $\log x$ on rectangular axes. Such a format is preferred when both variables contain data values that

span more than one order of magnitude. With data that follow a trend of the form $y = ax^n$, a linear curve will be obtained in log-log format.

Significant Digits

Significant digits refer to the number of digits found before and after the decimal point in a reported number. While leading zeros are not significant, all trailing zeros are significant. For example, 42.0 has three significant digits, including one decimal digit. The number of significant digits needed depends on the problem and the close discretion of the engineer. But in the course of working a problem, there should be consistency in the number of significant digits used and reported. The number of digits reported reflects a measure of the uncertainty in the value assigned by the engineer. So to determine the number of significant digits required, just ask yourself: “What value makes sense to this problem?” For example, to maintain a relative uncertainty of within 1%, we need to report values to three significant figures. But because rounding errors tend to accumulate during calculations, we would perform all intermediate calculations to at least four significant figures, and then round the final result down to three significant figures.

1.7 SUMMARY

During a measurement the input signal is not known but is inferred from the value of the output signal from the measurement system. We discussed the process of calibration as the means to relate a measured input value to the measurement system output value and the role of standards in that process. An important step in the design of a measurement system is the inclusion of a means for a reproducible calibration that closely simulates the type of signal to be input during actual measurements. A test is the process of “asking a question.” The idea of a test plan was developed to answer that question. However, a measured output signal can be affected by many variables that will introduce variation and trends and confuse that answer. Careful test planning is required to reduce such effects. A number of test plan strategies were developed, including randomization. The popular term “accuracy” was explained in terms of the more useful concepts of random error, systematic error, random uncertainty, and systematic uncertainty and their effects on a measured value. We also explored the idea of test standards and engineering codes, legal documents that influence practically every manufactured product around us.

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NOMENCLATURE

e	absolute error	u_R	repeatability uncertainty; uncertainty assigned to repeatability error
p	pressure ($ml^{-1}t^{-2}$)	u_z	zero uncertainty; uncertainty assigned to zero error
r_i	input span	x	independent variable; input value; measured variable
r_o	output span	y	dependent variable; output value
s_x	standard deviation of x	y_L	linear polynomial
u_c	overall instrument uncertainty	A	relative error; relative accuracy
u_h	hysteresis uncertainty; uncertainty assigned to hysteresis error	K	static sensitivity
u_K	sensitivity uncertainty; uncertainty assigned to sensitivity error	T	temperature ($^{\circ}$)
u_L	linearity uncertainty; uncertainty assigned to linearity error	$\forall$	volume (l^3)

PROBLEMS

- 1.1 Discuss your understanding of the hierarchy of standards beginning with the primary standard. In general, what is meant by the term “standard”? Can you cite examples of standards in everyday use?
- 1.2 What is the purpose of a calibration? Suppose an instrument is labeled as “calibrated.” What should this mean to you as an engineer?
- 1.3 Suppose you found a dial thermometer in a stockroom. Discuss several methods by which you might estimate random and systematic error in the thermometer? How would you estimate its uncertainty?
- 1.4 Consider the example described in Figure 1.6. Discuss the effect of the extraneous variable, barometric pressure, in terms of noise and interference relative to any one test and relative to several tests. Discuss how the interference effect can be broken up into noise.
- 1.5 How would the resolution of the display scale of an instrument affect its uncertainty? Suppose the scale was somehow offset by one least count of resolution. How would this affect its uncertainty? Explain in terms of random and systematic error.
- 1.6 How would the hysteresis of an instrument affect its uncertainty? Explain in terms of random and systematic error.

- 1.7** Select three different types of measurement systems with which you have experience, and identify which attributes of the system comprise the measurement system stages of Figure 1.5.
- 1.8** Identify the measurement system stages for the following systems (refer back to Figure 1.5 and use other resources, such as a library or Internet search, as needed to learn more about each system):
- Room thermostat
 - Automobile speedometer
 - Portable CD stereo system
 - Antilock braking system (automobile)
 - Automobile cruise control
 - Throttle position sensor for a Formula One race car
- 1.9** What is the range of the calibration data of Table 1.5?

Table 1.5 Calibration Data

X [cm]	Y [V]	X [cm]	Y [V]
0.5	0.4	10.0	15.8
1.0	1.0	20.0	36.4
2.0	2.3	50.0	110.1
5.0	6.9	100.0	253.2

- 1.10** For the calibration data of Table 1.5, plot the results using rectangular and log-log scales. Discuss the apparent advantages of either presentation.
- 1.11** For the calibration data of Table 1.5, determine the static sensitivity of the system at (a) $X = 5$; (b) $X = 10$; and (c) $X = 20$. For which input values is the system more sensitive? Explain what this might mean in terms of a measurement and in terms of measurement errors.
- 1.12** Consider the voltmeter calibration data in Table 1.6. Plot the data using a suitable scale. Specify the percent maximum hysteresis based on full-scale range. Input X is based on a standard known to be accurate to better than 0.05 mV.

Table 1.6 Voltmeter Calibration Data

Increasing Input [mV]		Decreasing Input [mV]	
X	Y	X	Y
0.0	0.1	5.0	5.0
1.0	1.1	4.0	4.2
2.0	2.1	3.0	3.2
3.0	3.0	2.0	2.2
4.0	4.1	1.0	1.2
5.0	5.0	0.0	0.2

- 1.13** Three clocks are compared to a time standard at three successive hours. The data are given in Table 1.7. Using Figure 1.12 as a guide, arrange these clocks in order of estimated accuracy. Discuss your choices.

Table 1.7 Clock Calibration Data

Clock	Standard Time		
	1:00:00	2:00:00	3:00:00
	Indicated Time		
A	1:02:23	2:02:24	3:02:25
B	1:00:05	2:00:05	3:00:05
C	1:00:01	1:59:58	3:00:01

1.14 Each of the following equations can be represented as a straight line on an x - y plot by choosing the appropriate axis scales. Plot them both in rectangular coordinate format and then in an appropriate format to yield a straight line. Explain how the plot operation yields the straight line. Variable y has units of volts. Variable x has units of meters (use a range of $0.01 < x < 10.0$). Note: This is easily done using a spreadsheet program where you can compare the use of different axis scales.

- a. $y = x^2$
- b. $y = 1.1x$
- c. $y = 2x^{0.5}$
- d. $y = 10x^4$
- e. $y = 10e^{-2x}$

1.15 Plot $y = 10e^{-5x}$ volts on in semilog format (use three cycles). Determine the slope of the equation at $x = 0$; $x = 2$; and $x = 20$.

1.16 Plot the following data on appropriate axes. Estimate the static sensitivity K at each X .

Y [V]	X [V]
2.9	0.5
3.5	1.0
4.7	2.0
9.0	5.0

1.17 The following data have the form $y = ax^b$. Plot the data in an appropriate format to estimate the coefficients a and b . Estimate the static sensitivity K at each value of X . How is K affected by X ?

Y [cm]	X [m]
0.14	0.5
2.51	2.0
15.30	5.0
63.71	10.0

1.18 For the calibration data given, plot the calibration curve using suitable axes. Estimate the static sensitivity of the system at each X . Then plot K against X . Comment on the behavior of the static sensitivity with static input magnitude for this system.

Y [cm]	X [kPa]
4.76	0.05
4.52	0.1
3.03	0.5
1.84	1.0

1.19 Consider the function

$$y = \frac{1}{x}$$

- a. Plot the function on rectangular coordinates.
 - b. Plot the function on semilog coordinates.
 - c. Plot the function on log-log coordinates.
 - d. Discuss the implications of each plot as it would relate to analyzing experimental data.
- 1.20** A bulb thermometer hangs outside a window and is used to measure the outside temperature. Comment on some extraneous variables that might affect the difference between the actual outside temperature and the indicated temperature from the thermometer.
- 1.21** A synchronous electric motor test stand permits either the variation of input voltage or output shaft load and the subsequent measurement of motor efficiency, winding temperature, and input current. Comment on the independent, dependent, and extraneous variables for a motor test.
- 1.22** The transducer specified in Table 1.1 is chosen to measure a nominal pressure of 500 cm H₂O. The ambient temperature is expected to vary between 18° and 25°C during tests. Estimate the magnitude of each elemental error affecting the measured pressure.
- 1.23** A force measurement system (weight scale) has the following specifications:

Range:	0 to 1000 N
Linearity error:	0.10% FSO
Hysteresis error:	0.10% FSO
Sensitivity error:	0.15% FSO
Zero drift:	0.20% FSO

Estimate the overall instrument uncertainty for this system based on available information. FSO refers to full-scale operating range.

- 1.24** An engineer ponders a test plan, thinking: “What strategy should I include to estimate any variation over time of the measured variable? What strategy should I include to estimate my control of the independent variable during the test?” What does the engineer mean?
- 1.25** If the outcome of a test was suspected to be dependent on the ability to control the test operating conditions, what strategy should be incorporated into the test plan to estimate this effect?
- 1.26** State the purpose of using randomization methods during a test. Develop an example to illustrate your point.
- 1.27** Provide an example of repetition and replication in a test plan from your own experience.
- 1.28** Develop a test plan that might be used to estimate the average temperature that could be maintained in a heated room as a function of the heater thermostat setting.

- 1.29** Develop a test plan that might be used to evaluate the fuel efficiency of a production model automobile. Explain your reasoning.
- 1.30** A race engine shop has just completed two engines of the same design. How might you determine which engine performs better: (a) on a test stand (engine dynamometer) or (b) at the race track? Describe some measurements that you feel might be useful and how you might use that information. Discuss possible differences between the two tests and how these might influence the results (e.g., you can control room conditions on a test stand but not at a track).
- 1.31** A large batch of carefully made machine shafts can be manufactured on 1 of 4 lathes by 1 of 12 quality machinists. Set up a test matrix to estimate the tolerances that can be held within a production batch. Explain your reasoning.
- 1.32** Suggest an approach(es) to estimate the linearity error and the hysteresis error of a measurement system.
- 1.33** Suggest a test matrix to evaluate the wear performance of four different brands of aftermarket passenger car tires of the same size, load, and speed ratings on a fleet of eight cars of the same make. If the cars were not of the same make, what would change?
- 1.34** The relation between the flow rate, Q , through a pipeline of area A and the pressure drop, Δp , across an orifice-type flow meter inserted in that line (Fig. 1.14) is given by

$$Q = CA\sqrt{\frac{2\Delta p}{\rho}}$$

where ρ is density and C is a coefficient. For a pipe diameter of 1 m and a flow range of 20°C water between 2 and 10 m³/min and $C = 0.75$, plot the expected form of the calibration curve for flow rate versus pressure drop over the flow range. Is the static sensitivity a constant? Incidentally, such an instrument test is described by ANSI/ASME Test Standard PTC 19.5.

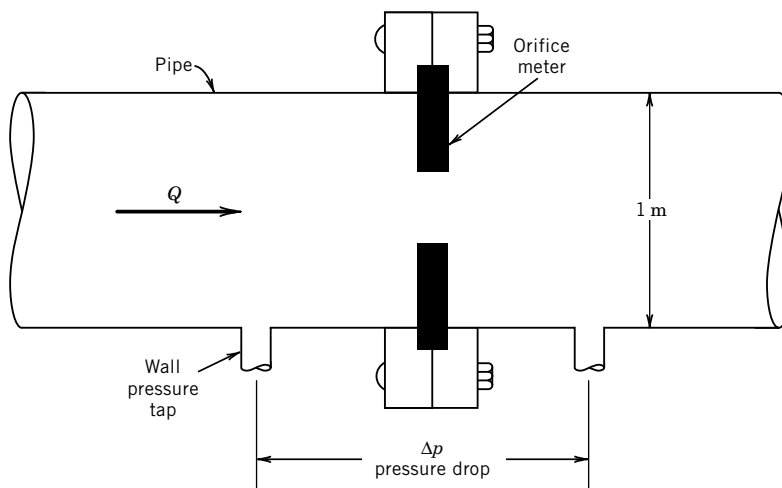


Figure 1.14 Orifice flow meter setup used for Problem 1.34.

- 1.35** The sale of motor fuel is an essential business in the global economy. Federal (US) law requires accurate measurement of the quantity of fuel delivered at a retail establishment. Determine:
- The federal standard for minimum accuracy of a gas pump.
 - The maximum allowable error in the delivery of 25 gallons of fuel (or 100 L of fuel).
 - An estimate of the cost of the maximum allowable error over the 10-year life of a vehicle that is driven 15,000 miles per year at the average fuel mileage of the American passenger car fleet.
- 1.36** For the orifice meter calibration in Problem 1.34: Would the term “linearity error” have a meaning for this system? Explain. Also, list the dependent and independent variables in the calibration.
- 1.37** A piston engine manufacturer uses four different subcontractors to plate the pistons for a make of engine. Plating thickness is important in quality control (performance and part life). Devise a test matrix to assess how well the manufacturer can control plating under its current system.
- 1.38** A simple thermocouple circuit is formed using two wires of different alloy: one end of the wires is twisted together to form the measuring junction, while the other ends are connected to a voltmeter and form the reference junction. A voltage is set up by the difference in temperature between the two junctions. For a given pair of alloy material and reference junction temperature, the temperature of the measuring junction is inferred from the measured voltage difference. For a measurement, what variables need to be controlled? What are the dependent and independent variables?
- 1.39** A linear variable displacement transducer (LVDT) senses displacement and indicates a voltage output, which is linear to the input. Figure 1.15 shows an LVDT setup used for static calibration. It uses a micrometer to apply the known displacement and a voltmeter for the output. A well-defined voltage powers the transducer. What are the independent and dependent variables in this calibration? Can you suggest any extraneous variables? What would be involved in a replication?
- 1.40** For the LVDT calibration of the previous problem, what would be involved in determining the repeatability of the instrument? The reproducibility? What effects are different in the two tests? Explain.

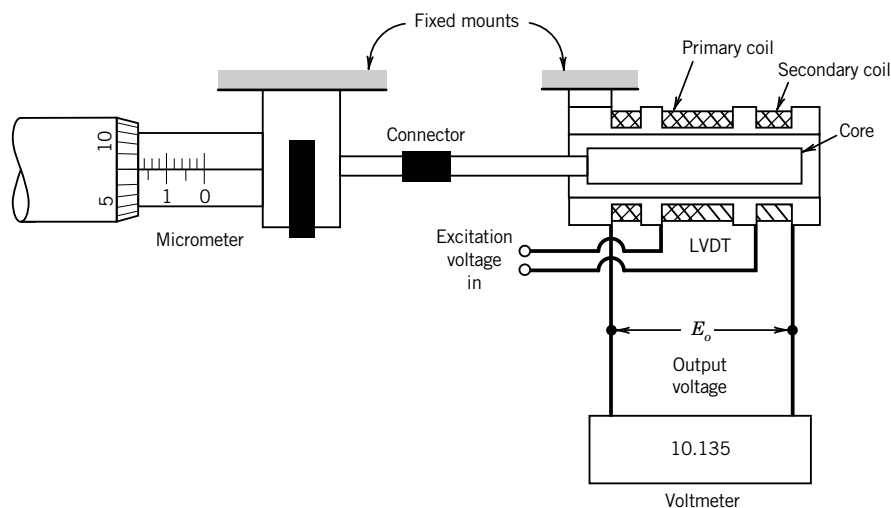


Figure 1.15 LVDT setup used for Problem 1.39.

- 1.41** A manufacturer wants to quantify the expected average fuel mileage of a product line of automobiles. They decide that they can put one or more cars on a chassis dynamometer and run the wheels at desired speeds and loads to assess this, or they can use drivers and drive the cars over some selected course instead. (a) Discuss the merits of either approach, considering the control of variables and identifying extraneous variables. (b) Can you recognize that somewhat different tests might provide answers to different questions? For example, discuss the difference in meanings possible from the results of operating one car on the dynamometer and comparing it to one driver on a course. Cite other examples. (c) Are these two test methods examples of concomitant methods?
- 1.42** We need to measure the coefficient of restitution of a volleyball by dropping the ball from a known height H and measuring the height of the bounce, h . The coefficient of restitution, C_R , is then calculated as

$$C_R = \sqrt{\frac{h}{H}}$$

- a. Develop a test plan for measuring C_R that includes the range of conditions expected in college-level volleyball play.
- b. Identify extraneous variables that may affect the resulting values of C_R .
- 1.43** Light gates may be used to measure the speed of projectiles, such as arrows shot from a bow. English long bows made of yew wood in the 1400s achieved launch speeds of 60 m/s. Determine the relationship between the distance between light gates and the accuracy required for sensing the times when the light gate senses the presence of the arrow.
- 1.44** You estimate your car's fuel use by comparing fuel volume used over a known distance. Your brother, who drives the same model car, disagrees with your claimed results based on his own experience. How might you justify the differences based on the concepts of control of variables, interference and noises effects, and test matrix used?
- 1.45** In discussing concomitant methods, we cited an example of computing the volume of a cylindrical rod based on its average dimensions versus known weight and material properties. While we should not expect too different an answer with either technique, identify where noise and interference effects will affect the result of either method.
- 1.46** When a strain gauge is stretched under uniaxial tension, its resistance varies with the imposed strain. A resistance bridge circuit is used to convert the resistance change into a voltage. Suppose a known tensile load were applied to the system shown in Figure 1.16 and the output measured on a voltmeter. What are the independent and dependent variables in this calibration? Can you suggest any extraneous variables? What would be involved in a replication?
- 1.47** For the strain gauge calibration of the previous problem, what would be involved in determining the repeatability of the instrument? The reproducibility? What effects are different in the tests? Explain.
- 1.48** A major tennis manufacturer is undertaking a test program for shoes, tennis balls, and tennis strings. Develop a test plan for the situations described below. In each case provide details for how the test should be conducted, and describe expected difficulties in interpreting the results. In each case the tests are to be conducted during college tennis matches. Four teams of six players each are to test the products under match (as opposed to practice) conditions. A tennis team consists of six players, and a match consists of six singles matches and three doubles matches.
- a. Tennis shoes: Two different sole designs and materials are to be wear tested. The life of the soles are known to be strongly affected by court surface and playing style.

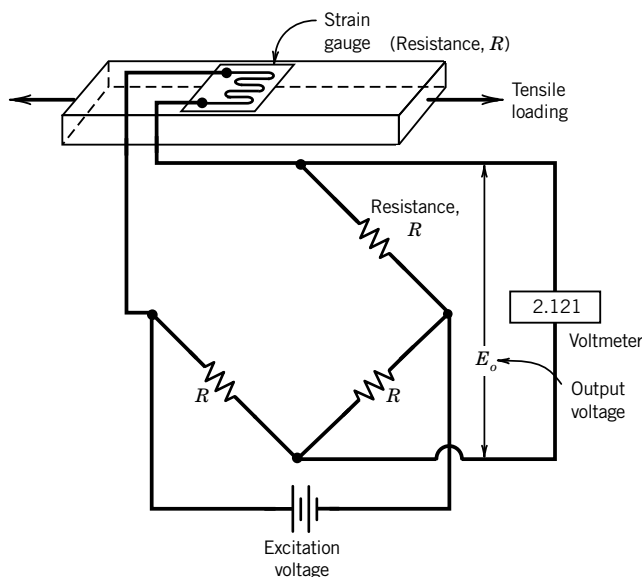


Figure 1.16 Strain gauge setup used for Problem 1.46.

- b. Tennis strings: A new tennis string material is to be tested for durability. String failure occurs due to breakage or due to loss of tension. String life is a function of the racquet, the player's style, and string tension.
 - c. Tennis balls: Two tennis balls are to be play tested for durability. The condition of a tennis ball can be described by the coefficient of restitution, and the total weight (since the cover material actually is lost during play). A player's style and the court surface are the primary determining factors on tennis ball wear.
- 1.49** The acceleration of a cart down a plane inclined at an angle α to horizontal can be determined by measuring the change in speed of the cart at two points, separated by a distance s , along the inclined plane. Suppose two photocells are fixed at the two points along the plane. Each photocell measures the time for the cart, which has a length L , to pass it. Identify the important variables in this test. List any assumptions that you feel are intrinsic to such a test. Suggest a concomitant approach. How would you interpret the data to answer the question?
- 1.50** Is it more fuel efficient to drive a car in warm, humid weather with the air conditioning on and windows closed, or with the air conditioning off and the windows open for ventilation? Develop a test plan to address this question. Include a concomitant approach (experimental or analytical) that might assess the validity of your test results.
- 1.51** Explain the potential differences in the following evaluations of an instrument's accuracy. Figure 1.12 will be useful, and you may refer to ASTM E177, if needed.
- a. The closeness of agreement between the true value and the average of a large set of measurements.
 - b. The closeness of agreement between the true value and an individual measurement.
- 1.52** Suggest a reasonable number of significant digits for reporting the following common values and give some indication as to your reasoning:
- a. Your body weight for a passport
 - b. A car's fuel usage (use liters per 100 km)

- c. The weight of a bar of pure (at least 99.5%) gold (consider a 1 kg_f bar and a 100 oz bar)
 - d. Distance traveled by a body in 1 second if moving at 1 m/s (use meters)
- 1.53** Research the following test codes (these are available in most libraries). Write a short (200-word) report that describes the intent and an overview of the code:
- a. ASTM F 558-88 (Air Performance of Vacuum Cleaners)
 - b. ANSI Z21.86 (Gas Fired Space Heating Appliances)
 - c. ISO 10770-1:1998 (Test Methods for Hydraulic Control Valves)
 - d. ANSI/ASME PTC19.1-2005 (Test Uncertainty)
 - e. ISO 7401:1988 (Road Vehicles: Lateral Response Test Methods)
 - f. Your local municipal building code or housing ordinance
 - g. Any other code assigned by your instructor
- 1.54** Show how the following functions can be transformed into a linear curve of the form $Y = a_1X + a_0$ where a_1 and a_0 are constants. Let m , b , and c be constants.
- a. $y = bx^m$
 - b. $y = be^{mx}$
 - c. $y = b + c\sqrt[m]{x}$